Mc Donald's Sales Data Analysis Dashboard

A comprehensive Excel-based analytics solution leveraging Power Query, Power Pivot, and Pivot Tables to transform raw sales data into actionable business insights

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Project Objectives

This project aims to analyze restaurant order data systematically to uncover valuable patterns and generate data-driven business insights

Revenue Analysis

Calculate total revenue generated from menu items and identify revenue contribution by different categories

Popular Items

Identify the most popular food items and analyze daily order trends across the business

Interactive Dashboard

Create an interactive dashboard with slicers and pivot tables for dynamic business insights

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Dataset Structure

The analysis leverages two interconnected tables containing comprehensive order and menu information

Orders Table

- Order ID (unique identifier)
- Order Date (transaction date)
- Order Time (time of day)
- Item ID (links to menu)

Menu Table

- Menu Item ID (unique)
- Item Name (product title)
- Category (food type)
- Price (individual item cost)

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Data Cleaning Process

Data quality was ensured through systematic cleaning using Excel's Power Query functionality

01

Duplicate Removal

Identified and eliminated duplicate records to ensure data integrity

02

Missing Values

Checked for and addressed missing values across all fields

03

Date Standardization

Standardized date format for consistent time-based analysis

04

Price Validation

Ensured numeric format for price column and verified data consistency

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Data Modeling with Power Pivot

The dataset was loaded into the Power Pivot Data Model, establishing relationships and creating calculated measures using DAX

Relationship Created

Orders[item_id] ↠ Menu[menu_item_id]

This established a one-to-many relationship enabling cross-table analysis

Calculated Measures

Total Revenue:

SUMX(Orders, RELATED(Menu[price]))

Additional measures included order counts, average values, and category aggregations

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Key Analysis Performed

Comprehensive analytical approach covering multiple dimensions of restaurant operations

1

Total Revenue Calculation

Summed all menu item prices across completed orders

2

Daily Orders Analysis

Tracked number of orders placed each day of the week

3

Popular Item Identification

Identified most frequently ordered menu items

4

Category Revenue

Analyzed revenue contribution by different menu categories

5

Average Items Per Order

Calculated mean number of items per customer transaction

6

Time of Day Patterns

Examined order volume distribution by time of day

7

Weekend vs Weekday

Compared sales performance between weekends and weekdays

8

Top Selling Items

Ranked menu items by frequency and revenue contribution

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Key Insights Discovered

61003.38

Total Revenue

Generated from analyzed order data

2.26

Average Items Per Order

Indicates typical customer purchase pattern

Top Performers

- **Burger category** generated highest revenue
- **Big Mac** was most frequently ordered item
- Peak order time: **12 PM – 2 PM**

Pattern Observations

Weekdays generated higher revenue compared to weekends, indicating weekday lunch rush drives primary sales volume

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Dashboard Features

Interactive Excel dashboard created using Pivot Tables, Pivot Charts, and Slicers for dynamic data exploration

Pivot Tables

Dynamic tables for revenue by category, orders by day, and top selling items analysis

Pivot Charts

Visual representations including bar charts, line graphs, and pie charts for intuitive understanding

Slicers

Interactive filters allowing users to drill down by date, category, and time period

KPI Cards

Summary cards displaying key metrics like total revenue, order count, and average ticket size

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Business Impact

This analytics solution delivers practical value for restaurant management and strategic decision-making

Performance Tracking

Enables management to track sales performance across multiple dimensions in real-time

Category Insights

Identifies high-performing menu categories and items requiring optimization

Menu Optimization

Supports data-driven decisions about menu offerings, pricing, and promotions

Better Decisions

Empowers management with insights for inventory planning, staffing, and marketing

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Conclusion

Successfully Analyzed

Restaurant sales data transformed into meaningful insights

Built Interactive

Dashboard with slicers and pivot tables for easy exploration

Extracted Insights

Valuable business intelligence for strategic decision-making

Excel Power

Demonstrated practical application of Excel for data analytics

This project showcases how Microsoft Excel, when leveraged with Power Query, Power Pivot, and Pivot Tables, can deliver professional-grade business intelligence solutions without requiring specialized software

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